



DAYANANDA SAGAR COLLEGE OF ENGINEERING

(An Autonomous Institution Approved by AICTE, Affiliated to VTU, Belagavi, Karnataka)

Shavige Malleshwara Hills, Kumaraswamy Layout Bengaluru – 560 111

Department of Computer Science & Engineering

22CS66-6th Sem Mini Project Phase-1- Report 2024-2025

TITLE OF THE PROJECT	AI-Powered Multidisciplinary Health Diagnosis System: A Unified Tool for Early Detection of Chronic Diseases		
STUDENT NAME	Divyam Saini	Krish Dipesh Makadia	Madhab Patwari
USN	1DS22CS079	1DS22CS110	1DS22CS113
INDIVIDUAL CONTRIBUTION	Diabetes Prediction and Streamlit Interface Design	Heart Disease Prediction and Streamlit Interface Design	Parkinson’s Disease Prediction, Carotid Artery Diagnosis and Streamlit Interface Design
DESIGN/ METHODOLOGY	The project implements a unified system leveraging AI and machine learning to detect and predict four major health conditions: Diabetes, Heart Disease, Parkinson’s Disease, and Carotid Artery Abnormalities. The design methodology integrates predictive analytics, image processing, and deep learning to provide accurate results through an interactive interface. 1. Overall System Design • Data-Driven Approach: Utilized pre-trained machine learning models for structured data (diabetes, heart disease, and Parkinson’s disease) and deep learning for unstructured image data (carotid artery diagnosis). • Modular Design: Each disease module is implemented as an independent unit within the unified platform for scalability and ease of updates. • User Interface: Developed a streamlined navigation system using Streamlit, enabling users to choose a specific condition and interact with the system effortlessly.		



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2. Disease-Specific Methodologies

Diabetes Prediction

- **Input Features:** Collected data such as pregnancies, glucose levels, blood pressure, insulin levels, BMI, age, and pedigree function.
- **Model:** A logistic regression classifier trained using the **PIMA Indian Diabetes Dataset**.
- **Pipeline:**
 - Preprocessed user inputs to match training data format.
 - Predicted outcome using the serialized model (pickle).
 - Displayed diagnosis with contextual information.

Heart Disease Prediction

- **Input Features:** Used clinical features such as age, sex, chest pain type, resting blood pressure, cholesterol levels, fasting blood sugar, electrocardiogram results, and exercise-induced metrics.
- **Model:** Decision tree classifier trained on the **Cleveland Heart Disease Dataset**.
- **Pipeline:**
 - Accepted user inputs in a structured format.
 - Predicted heart disease probability with the trained model.
 - Provided interpretive feedback to the user.

Parkinson's Disease Prediction

- **Input Features:** Acoustic features including jitter, shimmer, and harmonic-to-noise ratio from voice samples.
 - **Model:** A support vector machine (SVM) classifier trained on the **Parkinson's Disease Voice Dataset**.
 - **Pipeline:**
 - Processed user input features into a compatible format.
 - Predicted the presence of Parkinson's disease using the trained model.
 - Delivered the diagnosis with an explanatory note for clarity.
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Carotid Artery Diagnosis

- **Input Data:** Ultrasound images of carotid arteries uploaded by the user.
- **Model:** A UNet-based Convolutional Neural Network (CNN) trained for semantic segmentation.
- **Pipeline:**
 - **Preprocessing:** Resized images, normalized pixel values, and converted to tensor format.
 - **Prediction:** Generated segmentation mask for the carotid artery.
 - **Postprocessing:**
 - Extracted contours to highlight artery boundaries.
 - Overlaid segmentation mask on the original image for visual context.
 - Extracted segmented region for focused analysis.
 - Displayed results, including the original image, segmentation overlay, and segmented region.

3. User Interface (UI) Design

- **Streamlit Framework:** Created a responsive web-based interface.
- **Navigation:** Sidebar menu enables users to select specific disease modules.
- **Data Input:** Designed forms for text-based input and image upload functionality.
- **Results Visualization:** Incorporated success messages and dynamic image rendering for better interpretability.

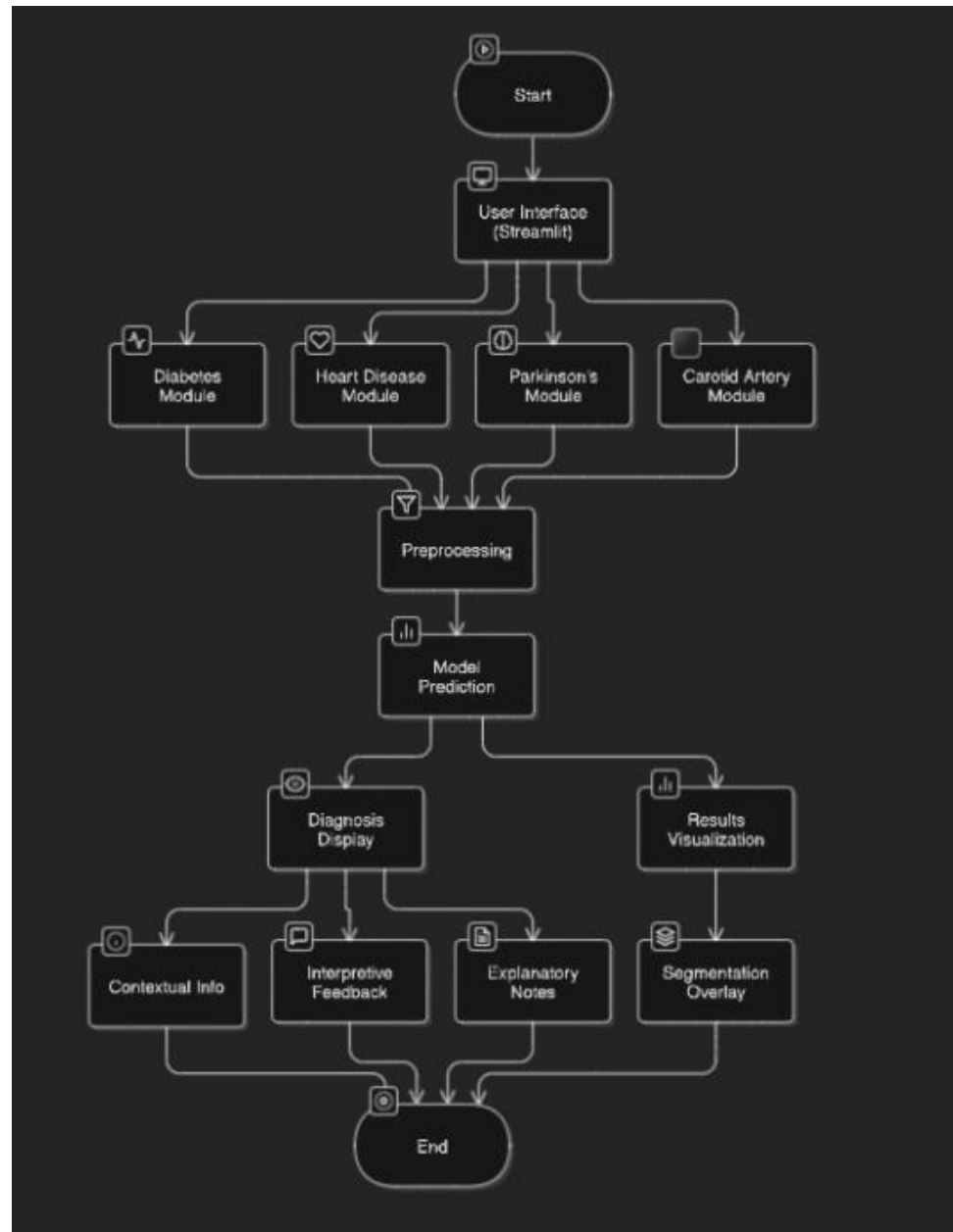


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PLATFORM USED

- ☐ **Programming Language:** Python
- ☐ **Frameworks/Libraries:** Streamlit, Keras, scikit-learn, OpenCV, NumPy, Matplotlib
- ☐ **Models:** Logistic Regression, Decision Tree, SVM, and UNet-based CNN
- ☐ **Deployment:** Streamlit platform for real-time, web-accessible predictions.



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PROJECT CODE LINK (GITHUB/ GOOGLE DRIVE)	https://github.com/Madhab2101/Multiple-Disease-Prediction.git
CONCLUSION	The Multiple Disease Prediction System successfully integrates multiple disease detection models into a single user-friendly platform. By leveraging advanced machine learning and deep learning techniques, it provides accurate predictions for diabetes, heart disease, Parkinson's disease, and carotid artery conditions. This innovative system demonstrates the potential of AI in revolutionizing healthcare by enabling early diagnosis, improving accessibility, and empowering individuals to take proactive measures in managing their health. Future enhancements, such as incorporating additional disease models and refining prediction algorithms, can further expand the system's utility. This project highlights the transformative role of AI in creating a unified and efficient healthcare ecosystem.



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SCREENSHOTS/ RESULTS

Smart Disease
Detection Hub

Carotid Artery Diagnosis

Diabetes Prediction

Heart Disease Prediction

Parkinsons Prediction

Smart Disease
Detection Hub

Carotid Artery Diagnosis

Heart Disease Prediction

Parkinsons Prediction

Smart Disease
Detection Hub

Carotid Artery Diagnosis

Diabetes Prediction

Heart Disease Prediction

Parkinsons Prediction

Diabetes Prediction

Deploy

Number of Pregnancies

Glucose Level

Blood Pressure value

Skin Thickness value

Insulin Level

BMI value

Diabetes Pedigree Function value

Age of the Person

Diabetes Test Result

Heart Disease Prediction

Deploy

Age

Sex

Chest Pain Types

Resting Blood Pressure

Serum Cholesterol in mg/dl

Fasting Blood Sugar > 120 mg/dl

Resting Electrocardiographic results

Maximum Heart Rate achieved

Exercise Induced Angina

ST depression induced by exercise

Slope of the peak exercise ST segment

Major vessels colored by fluoroscopy

thal: 0 = normal; 1 = fixed defect; 2 = reversible defect

Heart Disease Test Result

Parkinson's Disease Prediction

Deploy

MDVP:F0(Hz)

MDVP:F1(Hz)

MDVP:F1o(Hz)

MDVP:jitter(%)

MDVP:jitter(Abs)

MDVP:RAP

MDVP:RPQ

Jitter:DDP

MDVP:Shimmer

MDVP:Shimmer(dB)

Shimmer:APQ3

Shimmer:APQ5

MDVP:APQ

Shimmer:DDA

NHR

HNR

RPDE

DFA

spread1

spread2

D2

PPE

Parkinson's Test Result



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Smart Disease Detection Hub

Carotid Artery Diagnosis

Diabetes Prediction

Heart Disease Prediction

Parkinsons Prediction

Deploy

Carotid Artery Diagnosis

Upload an Ultrasound image of the Carotid Artery

Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG, BMP, TIFF, TIF

Browse files

Smart Disease Detection Hub

Carotid Artery Diagnosis

Diabetes Prediction

Heart Disease Prediction

Parkinsons Prediction

Deploy

Diabetes Prediction

Number of Pregnancies	Glucose Level	Blood Pressure value
6	134	70
Skin Thickness value	Insulin Level	BMI value
23	130	35.4
Diabetes Pedigree Function value	Age of the Person	
0.542	29	

Diabetes Test Result

The person is diabetic

Smart Disease Detection Hub

Carotid Artery Diagnosis

Diabetes Prediction

Heart Disease Prediction

Parkinsons Prediction

Deploy

Heart Disease Prediction

Age	Sex	Chest Pain types
63	1	3
Resting Blood Pressure	Serum Cholesterol in mg/dl	Fasting Blood Sugar > 120 mg/dl
145	233	1
Resting Electrocardiographic results	Maximum Heart Rate achieved	Exercise Induced Angina
0	150	0
ST depression induced by exercise	Slope of the peak exercise ST segment	Major vessels colored by fluoroscopy
2.3	0	0

thal: 0 = normal; 1 = fixed defect; 2 = reversible defect

Heart Disease Test Result

The person is having heart disease



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Smart Disease Detection Hub

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Parkinsons Prediction

Smart Disease Detection Hub

Carotid Artery Diagnosis

Diabetes Prediction

Heart Disease Prediction

Parkinsons Prediction

Smart Disease Detection Hub

Carotid Artery Diagnosis

Diabetes Prediction

Heart Disease Prediction

Parkinsons Prediction

Parkinson's Disease Prediction

MDVP:F0(Hz)	MDVP:F1(Hz)	MDVP:F2(Hz)	MDVP:Jitter(%)	MDVP:Jitter(Abs)
119.99200	157.30200	74.99700	0.00784	0.00007
MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer(dB)
0.00370	0.00554	0.01109	0.04374	0.42600
Shimmer:APQ3	Shimmer:APQ5	MDVP:APQ	Shimmer:DDA	NHR
0.02182	0.03130	0.02971	0.06545	0.02211
HNR	RPDE	DFA	spread1	spread2
21.03300	0.414783	0.815285	-4.813031	0.266482
D2	PPE			
2.301442	0.284654			

Parkinson's Test Result

The person has Parkinson's disease

Carotid Artery Diagnosis

Upload an Ultrasound Image of the Carotid Artery

Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG, BMP, TIFF, TIF

202202071357530054VAS_slice_914.png 308.6KB

Uploaded Image

Analyze Carotid Artery

Original Ultrasound Image

Predicted Mask with Boundary

Extracted Segmented Region

Carotid Artery Analysis Completed